Pell equations

L4-X4 Level 4 Olympiad · Number theory · Beyond the textbook

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The equation x^2 - dy^2 = 1 and its relatives, covering the fundamental solution, the descent that generates every solution, the negative Pell equation, classes of solutions of x^2 - dy^2 = n, and how contest problems hide them.
By the end you can find the fundamental solution of a Pell equation and prove that its powers give every solution; solve x^2 - dy^2 = n completely by finding its classes of solutions; recognise a Pell equation hidden in a contest problem and handle the side conditions.
Prerequisites: Olympiad Diophantine equations (L4-NT2). **Contests:** BMO Round 1, BMO Round 2, AIME, USAJMO, USAMO. **Time:** about 120 minutes.

1 Lesson

Find all positive integers n for which $3n+1$ and $4n+1$ are both perfect squares.

A short search finds $n = 56$, with $3n+1 = 13^2$ and $4n+1 = 15^2$, and nothing else below 10 000. Writing $3n+1 = a^2$ and $4n+1 = b^2$ and eliminating n gives $4a^2 - 3b^2 = 1$. This does not factorise over the integers, nothing bounds a and b, and no modulus rules the equation out, so the methods of L4-NT2 do not finish the problem. There are infinitely many solutions, and the next one is $n = 10\,920$.

The equation $x^2 - dy^2 = 1$ is the **Pell equation**. Its theory is old. Brahmagupta (628) knew how to combine solutions, Bhaskara II (1150) gave an algorithm for finding one, Fermat challenged the English mathematicians with $x^2 - 61y^2 = 1$ in 1657, and Lagrange proved in 1768 that a solution always exists. Euler attached Pell's name by mistake. School courses skip the equation because every Diophantine equation met there has finitely many solutions. A contest problem that hides a Pell equation is exactly one where that fails, and the standard methods can find the small solutions but cannot prove that the list is complete.

Throughout, d is a positive integer that is not a perfect square, so $\sqrt{d}$ is irrational. For $u = x + y\sqrt{d}$ with x, y integers we write $\bar{u} = x - y\sqrt{d}$ and $N(u) = u\bar{u} = x^2 - dy^2$, the *norm* of u. Since $\sqrt{d}$ is irrational, u determines x and y.

1.1 Multiplying solutions

For all integers x_1, y_1, x_2, y_2,

$$(x_1^2 - dy_1^2)(x_2^2 - dy_2^2) = (x_1x_2 + dy_1y_2)^2 - d(x_1y_2 + x_2y_1)^2.$$

This is Brahmagupta's identity. In the notation above this identity says $N(uv) = N(u)N(v)$, because $(x_1 + y_1\sqrt{d})(x_2 + y_2\sqrt{d}) = (x_1x_2 + dy_1y_2) + (x_1y_2 + x_2y_1)\sqrt{d}$ and conjugation respects products. In particular, if $N(u) = 1$ then $\bar{u} = 1/u$, and every power u^k with $k \in \mathbb{Z}$ has integer coordinates and norm 1. So one solution of $x^2 - dy^2 = 1$ with $y \neq 0$ produces infinitely many. For $d = 3$, $(2+\sqrt{3})^2 = 7 + 4\sqrt{3}$ and $(2+\sqrt{3})^3 = 26 + 15\sqrt{3}$, and indeed $26^2 - 3 \cdot 15^2 = 1$.

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1.2 Existence and the fundamental solution

Theorem 1 (Lagrange).
The equation $x^2 - dy^2 = 1$ has a solution in positive integers.

Proof sketch. For each N, two of the $N+1$ fractional parts $\{k\sqrt{d}\}$, $0 \le k \le N$, lie in the same interval $[\frac{j}{N}, \frac{j+1}{N})$, and subtracting gives integers x and $1 \le y \le N$ with $|x - y\sqrt{d}| < \frac{1}{N} \le \frac{1}{y}$. Then $x + y\sqrt{d} < 2y\sqrt{d} + 1$, so $|x^2 - dy^2| < 2\sqrt{d} + 1$. Since $x - y\sqrt{d}$ is never 0, letting N grow gives infinitely many such pairs. By pigeonhole, infinitely many share the same value $k = x^2 - dy^2$ (a non-zero integer) and the same residues of x and y modulo $|k|$. Take two of them, $u_1 = x_1 + y_1\sqrt{d}$ and $u_2 = x_2 + y_2\sqrt{d}$ with $u_1 \ne u_2$. Then $u_1\bar{u}_2 = A + B\sqrt{d}$ with $A = x_1x_2 - dy_1y_2 \equiv x_1^2 - dy_1^2 \equiv 0$ and $B = x_2y_1 - x_1y_2 \equiv 0 \pmod{|k|}$, and $N(u_1\bar{u}_2) = k^2$. So $(A/k)^2 - d(B/k)^2 = 1$ in integers. If $B = 0$, then $u_1 = \frac{A}{k}u_2$ and comparing norms gives $\frac{A}{k} = \pm 1$, which is impossible as $u_1 \ne u_2$ and both are positive. So $(|A/k|, |B/k|)$ is a positive solution. □

Among the positive solutions, the one with the smallest y is the **fundamental solution** (x_1, y_1), and we write $\varepsilon = x_1 + y_1\sqrt{d}$. For positive solutions $x = \sqrt{1 + dy^2}$ increases with y, so ε is also the smallest number greater than 1 of the form $x + y\sqrt{d}$ with x, y positive and norm 1. For small d, try $y = 1, 2, 3, \ldots$ until $dy^2 + 1$ is a square. The values $d = 2, 3, 5, 6, 7, 8$ give $(3,2)$, $(2,1)$, $(9,4)$, $(5,2)$, $(8,3)$, $(3,1)$. Near a square there are formulas, as $d = m^2 - 1$ has fundamental solution $(m, 1)$, and $d = m^2 \pm 2$ has the solution $(m^2 \pm 1, m)$. Elsewhere the fundamental solution can be huge. For $d = 13$ the solution is $(649, 180)$ and for $d = 61$ the solution is $(1766319049, 226153980)$. Continued fractions find the fundamental solution in general, but contest problems choose d so that a search finds the solution.

1.3 All solutions and the descent

Theorem 2.
The positive solutions of $x^2 - dy^2 = 1$ are exactly the pairs (x_k, y_k), $k \ge 1$, given by $x_k + y_k\sqrt{d} = \varepsilon^k$. All integer solutions are given by $x + y\sqrt{d} = \pm\varepsilon^k$ with $k \in \mathbb{Z}$.

Proof. Let (x, y) be a positive solution and $u = x + y\sqrt{d}$. Then $u \ge \varepsilon > 1$, so there is $k \ge 1$ with $\varepsilon^k \le u < \varepsilon^{k+1}$. Put $v = u\varepsilon^{-k} = u\bar{\varepsilon}^k$. Since $\bar{\varepsilon} = x_1 - y_1\sqrt{d}$ has integer coordinates, $v = a + b\sqrt{d}$ with a, b integers, $N(v) = N(u)N(\bar{\varepsilon})^k = 1$, and $1 \le v < \varepsilon$. Suppose $v > 1$. Then $\bar{v} = 1/v$ lies in $(0, 1)$, so $a = \frac{v+\bar{v}}{2} > 0$ and $b\sqrt{d} = \frac{v-\bar{v}}{2} > 0$. So (a, b) is a positive solution with $a + b\sqrt{d} < \varepsilon$, contradicting the minimality of ε. Hence $v = 1$ and $u = \varepsilon^k$. For an arbitrary solution, $x + y\sqrt{d}$ is one of $\pm u, \pm\bar{u} = \pm u^{-1}$ for a positive solution u, or ± 1 (when $y = 0$). □

This is a descent in the sense of L4-NT2, since dividing by ε turns a solution into a smaller one, and the process can only stop at 1. A marker wants to see the three facts that make the argument work, which are that v has integer coordinates, v has norm 1, and $1 \le v < \varepsilon$ forces positive coordinates.

Comparing coefficients in $\varepsilon^{k+1} = \varepsilon^k\varepsilon$ gives $x_{k+1} = x_1x_k + dy_1y_k$ and $y_{k+1} = y_1x_k + x_1y_k$. Also $\varepsilon + \bar{\varepsilon} = 2x_1$ and $\varepsilon\bar{\varepsilon} = 1$, so both sequences satisfy $s_{k+1} = 2x_1s_k - s_{k-1}$, and $x_k = \frac{\varepsilon^k + \bar{\varepsilon}^k}{2}$, $y_k = \frac{\varepsilon^k - \bar{\varepsilon}^k}{2\sqrt{d}}$. Because these recurrences can be run backwards, the solutions are periodic modulo any m, which is how parity and divisibility side conditions are handled.

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Example 1.
Find all positive integers n for which $3n+1$ and $4n+1$ are both perfect squares.

Solution.
Write $3n+1=a^2$ and $4n+1=b^2$ with a,b positive. Then $4a^2-3b^2=1$, that is $X^2-3b^2=1$ with $X=2a$. Conversely, if $X^2-3b^2=1$ with X even, then $3b^2=X^2-1\equiv 3 \pmod 4$, so b is odd, $n=\frac{b^2-1}{4}$ is an integer, $4n+1=b^2$ and $3n+1=\frac{3b^2+1}{4}=a^2$. So we need the solutions of $X^2-3Y^2=1$ with X even.
The fundamental solution is $(2,1)$, so by Theorem 2 the positive solutions are $X_k+Y_k\sqrt{3}=(2+\sqrt{3})^k$, with $X_{k+1}=2X_k+3Y_k$ and $Y_{k+1}=X_k+2Y_k$. Modulo 2 this reads $(X_{k+1},Y_{k+1})\equiv(Y_k,X_k)$, and $(X_1,Y_1)\equiv(0,1)$, so X_k is even exactly when k is odd. The odd powers are $(2,1)$, $(26,15)$, $(362,209)$, $(5042,2911)$, and so on, and the first gives $n=0$. Hence the answers are $n=\frac{Y^2-1}{4}$ for $Y=15,209,2911,\ldots$, where Y runs through the sequence $1,15,209,\ldots$ with $Y_{j+1}=14Y_j-Y_{j-1}$ (the trace of $(2+\sqrt{3})^2=7+4\sqrt{3}$ is 14). The first three values are $n=56$, 10 920 and 2 118 480.
Remark. The search finds 56, and the factorisation and modular methods of L4-NT2 say nothing about completeness. Theorem 2 is what proves that there are no other solutions.

1.4 The negative Pell equation

The equation $x^2-dy^2=-1$ may or may not have solutions. If the equation has a solution, then $x^2\equiv -1$ modulo every prime factor of d and modulo 4 if $4\mid d$, so $4\nmid d$ and every odd prime factor of d is 1 mod 4. The converse fails, as $d=34$ passes this test and has no solutions (Practice 5).

Theorem 3.
Let η be the smallest number greater than 1 of the form $x+y\sqrt{d}$ with x,y integers and $x^2-dy^2=\pm 1$. Every such number that is positive is η^k for some $k\in\mathbb{Z}$. If $N(\eta)=1$, then $\eta=\varepsilon$ and $x^2-dy^2=-1$ has no solutions. If $N(\eta)=-1$, then $\varepsilon=\eta^2$, and the positive solutions of $x^2-dy^2=-1$ are given by the odd powers $\eta,\eta^3,\eta^5,\ldots$

Proof. First, η exists, because a number $u=x+y\sqrt{d}>1$ with $N(u)=\pm 1$ has $|\bar{u}|=1/u<1$, so $x=\frac{u+\bar{u}}{2}>0$ and $y\sqrt{d}=\frac{u-\bar{u}}{2}>0$, and there are finitely many such u below any bound. Now repeat the proof of Theorem 2 with η in place of ε. For such a u, choose k with $\eta^k\le u<\eta^{k+1}$, and then $v=u\eta^{-k}$ has integer coordinates, norm ± 1 and $1\le v<\eta$, and the argument just given shows that $v>1$ would make v a smaller candidate than η. So $u=\eta^k$, and $N(\eta^k)=N(\eta)^k$ gives the rest. □

For $d=2$, $\eta=1+\sqrt{2}$, so the solutions of $x^2-2y^2=-1$ are $(1,1)$, $(7,5)$, $(41,29)$, …, and $\varepsilon=(1+\sqrt{2})^2=3+2\sqrt{2}$. For $d=m^2+1$, $\eta=m+\sqrt{d}$.

1.5 The general equation $x^2-dy^2=n$

Multiplying a solution of $x^2-dy^2=n$ by ε gives another, so solutions come in *classes* $\{v\varepsilon^k : k\in\mathbb{Z}\}$. There may be several classes, or none, and the theorem below finds them all by a finite search.

Theorem 4.
Let $n\neq 0$. Every $u=x+y\sqrt{d}$ with x,y integers, $N(u)=n$ and $u>0$ can be written as $u=v\varepsilon^k$ with $k\in\mathbb{Z}$ and $v=a+b\sqrt{d}$ a solution with $\sqrt{|n|}\le v<\varepsilon\sqrt{|n|}$. Every such v

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satisfies $|a| < \frac{(\varepsilon+1)\sqrt{|n|}}{2}$ and $|b| < \frac{(\varepsilon+1)\sqrt{|n|}}{2\sqrt{d}}$. Moreover x and y are both positive if and only if $u > \sqrt{|n|}$.

Proof. Choose k with $\varepsilon^k\sqrt{|n|} \le u < \varepsilon^{k+1}\sqrt{|n|}$ and put $v = u\varepsilon^{-k}$. As before v has integer coordinates and $N(v) = n$. From $|\bar{v}| = |n|/v \le \sqrt{|n|}$ we get $|a| = \frac{|v+\bar{v}|}{2}$ and $|b|\sqrt{d} = \frac{|v-\bar{v}|}{2}$, both less than $\frac{\varepsilon\sqrt{|n|}+\sqrt{|n|}}{2}$. Finally, $x = \frac{u+\bar{u}}{2}$ and $y\sqrt{d} = \frac{u-\bar{u}}{2}$ are both positive if and only if $u > |\bar{u}| = |n|/u$, that is $u > \sqrt{|n|}$. □

So the positive solutions are the numbers $v\varepsilon^k$ with $k \ge 0$ and v in the window, except v itself when $v = \sqrt{|n|}$.

Example 2.
Find all positive integers y for which $2y^2 + 7$ is a perfect square.

Solution.
We need the positive solutions of $x^2 - 2y^2 = 7$. Here $\varepsilon = 3 + 2\sqrt{2} \approx 5.83$, and the window is $\sqrt{7} \le a + b\sqrt{2} < (3 + 2\sqrt{2})\sqrt{7}$, roughly $[2.65, 15.42)$. Theorem 4 gives $|a| \le 9$ and $|b| \le 6$. Checking $b = 0, \pm1, \ldots, \pm6$, the only solutions of $a^2 = 2b^2 + 7$ in this range are $(a, b) = (\pm3, \pm1)$ and $(\pm5, \pm3)$, and of the eight numbers $a + b\sqrt{2}$ only $3 + \sqrt{2} \approx 4.41$ and $5 + 3\sqrt{2} \approx 9.24$ lie in the window. So there are two classes, and the positive solutions are $(3 + \sqrt{2})\varepsilon^k$ and $(5 + 3\sqrt{2})\varepsilon^k$, $k \ge 0$:

$$3 + \sqrt{2},\ 13 + 9\sqrt{2},\ 75 + 53\sqrt{2},\ \ldots \qquad 5 + 3\sqrt{2},\ 27 + 19\sqrt{2},\ 157 + 111\sqrt{2},\ \ldots$$

The values of y are $1, 9, 53, 309, \ldots$ and $3, 19, 111, \ldots$, each sequence satisfying $y_{k+1} = 6y_k - y_{k-1}$.
Remark. The two classes are conjugate, as $5 + 3\sqrt{2} = (3 - \sqrt{2})\varepsilon$. Students often find the first class and stop. Without the window argument there is no proof that the list is complete.

1.6 Pell equations in disguise

Any equation $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ whose discriminant $B^2 - 4AC$ is positive and not a square becomes $X^2 - dY^2 = n$ after completing the square, with X and Y linear in x and y. The cost is a congruence condition on X and Y (parity, or a residue modulo $2A$), which is handled with the periodicity noted after Theorem 2. Vieta jumping (replacing one root of a quadratic in a by the other root) is the same descent, since after completing the square the step is multiplication by a unit of norm 1.

Example 3.
In a room of a people there are three times as many handshakes as in a room of b people, where $a, b \ge 2$ and every pair of people in a room shakes hands once. Find all possible pairs (a, b).

Solution.
The condition is $a(a-1) = 3b(b-1)$. Since a and $a-1$ are coprime, one could split the factors of $3b(b-1)$ between them, but that produces several cases, each an equation of the same type as the original. Instead complete the square, which gives $4a^2 - 4a = 12b^2 - 12b$ gives $(2a-1)^2 - 1 = 3\big((2b-1)^2 - 1\big)$, that is

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$$X^2 - 3Y^2 = -2, \qquad X = 2a - 1,\ Y = 2b - 1.$$

Conversely every solution with X, Y positive and odd gives a pair. Every solution of $X^2 - 3Y^2 = -2$ has X and Y odd, because if both were even the left side would be divisible by 4, and if exactly one were even the left side would be odd. So we need all positive solutions. With $\varepsilon = 2 + \sqrt{3}$ the window is $[\sqrt{2}, (2+\sqrt{3})\sqrt{2}) \approx [1.41, 5.28)$, and Theorem 4 gives $|X| \le 3$, $|Y| \le 1$. The solutions there are $(\pm 1, \pm 1)$, and only $1 + \sqrt{3}$ lies in the window. So there is one class, and the positive solutions are $(1+\sqrt{3})(2+\sqrt{3})^k$, $k \ge 0$:

$$1 + \sqrt{3},\ 5 + 3\sqrt{3},\ 19 + 11\sqrt{3},\ 71 + 41\sqrt{3},\ 265 + 153\sqrt{3},\ \ldots$$

The first gives $a = b = 1$, which is excluded. The rest give $(a, b) = (3, 2), (10, 6), (36, 21), (133, 77), \ldots$, and both a and b satisfy $s_{k+1} = 4s_k - s_{k-1} - 1$, because X and Y satisfy $s_{k+1} = 4s_k - s_{k-1}$. For instance $\binom{36}{2} = 630 = 3\binom{21}{2}$.

Remark. Vieta jumping is no help here, because as a quadratic in a, the other root of $a^2 - a - 3b(b-1) = 0$ is $1 - a$, and similarly for b, because the equation has no ab term. Completing the square supplies the unit $2 + \sqrt{3}$, which moves both variables at once.

Common traps.

- If d is a perfect square, $x^2 - dy^2$ factorises and there are finitely many solutions, so none of the theory applies. Check that d is not a square after completing the square.
- Finding the solutions is not solving the problem. A complete answer proves that the list is complete, using Theorem 2 for $n = \pm 1$ and the window of Theorem 4 for other n.
- For $n \ne \pm 1$ there can be several classes (Example 2). Check every v in the window, including those with negative coordinates.
- Side conditions (parity, divisibility by 3, $X = 2a - 1$) select a subsequence. Prove which one by periodicity of the recurrence, not by looking at three terms.
- The necessary conditions for $x^2 - dy^2 = -1$ are not sufficient, and $d = 34$ is the standard counterexample.
- Do not forget the trivial solutions $(\pm 1, 0)$, the negative solutions, and whether the problem asks for positive or non-negative values.

1.7 How to spot this in a contest

1. Two quadratic (or linear) expressions in one variable n must both be squares. Eliminate n.
2. Two figurate numbers are equal, or one is a fixed multiple of another. Typical cases are triangular numbers, $\binom{a}{2}$ and $\binom{b}{2}$, squares, sums of consecutive squares or cubes.
3. A quadratic equation in two variables whose discriminant is positive and not a square. Complete the square.
4. The problem asks you to prove that infinitely many integers have a quadratic property, or to find the largest or the k-th solution.
5. A sequence with $u_{k+1} = cu_k - u_{k-1}$, or with terms such as $\sqrt{5y^2 - 4}$, or a pair (a, b) with

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$ab \mid a^2 + b^2 + c$. This is descent on a Pell-type equation.

Once the equation is in the form $X^2 - dY^2 = n$, the work is always the same. Find ε, find the classes with Theorem 4, then impose the side conditions with the recurrence.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆ Find the fundamental solution of $x^2 - 11y^2 = 1$, and the next two solutions in positive integers.

Problem 2 ⋆⋆ Find the four smallest positive integers n for which $n+1$ and $5n+1$ are both perfect squares.

Problem 3 ⋆⋆ Prove that there are infinitely many positive integers n such that $n^2 + n - 1$ is five times a perfect square.

Problem 4 ⋆⋆⋆ Find all positive integers y less than 1000 for which $3y^2 + 13$ is a perfect square.

Problem 5 ⋆⋆⋆ Prove that the equation $x^2 - 34y^2 = -1$ has no solutions in integers, even though the congruence $x^2 \equiv -1 \pmod{34}$ has solutions.

Problem 6 ⋆⋆⋆⋆ Let (x_k, y_k), $k = 1, 2, 3, \ldots$, be the solutions of $x^2 - 2y^2 = 1$ in positive integers, in increasing order, so $(x_1, y_1) = (3, 2)$ and $(x_2, y_2) = (17, 12)$. Prove that $\gcd(y_m, y_n) = y_{\gcd(m,n)}$ for all $m, n \geq 1$, and find the smallest n for which 200 divides y_n.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AIME II 2008 Problem 15** ⋆⋆
   A difference of consecutive cubes equal to a square is a Pell equation after completing the square, and the second condition picks out the solutions.

2. **HMMT February 2017 Guts Round Problem 27** ⋆⋆
   Searching for the fundamental solution directly is slow, but 29 is close to a square, so look at the negative equation first.

3. **BMO2 2017/18 Problem 3** ⋆⋆
   Find the sum in closed form, and a quadratic factor must be a square. Complete the square and study the resulting Pell-type equation modulo a small number.

4. **HMMT February 2020 Guts Round Problem 21** ⋆⋆
   Show that a quadratic expression in consecutive terms is invariant, so the sequence obeys a linear recurrence of the form a(n+1) = c a(n) - a(n-1).

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5. **BMO2 2001/02 Problem 3** ⋆⋆⋆
   Show that $5y^2 - 4$ is a square for every term, and write the step from one term to the next as multiplication by a unit.

6. **IMO 1981 Problem 3** ⋆⋆⋆
   Show that one solution produces a smaller one, and descend.

7. **HMMT February 2017 Algebra and Number Theory Problem 8** ⋆⋆⋆
   Fix the value of the quotient and jump between the two roots of the quadratic to find all solutions from the smallest ones.

8. **BMO2 2006/07 Problem 2** ⋆⋆⋆
   Clear denominators and treat the equation as a quadratic in one variable, since the other root gives a new solution.

9. **IMO 1988 Problem 6** ⋆⋆⋆⋆
   Fix the quotient and descend from a minimal solution using the other root of the quadratic.

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4 Hints

1. Try $y = 1, 2, 3$ in turn. Then square and cube $x_1 + y_1\sqrt{11}$.
2. Eliminate n to get $b^2 - 5a^2 = -4$, and find its classes with Theorem 4.
3. Multiply by 4 and complete the square. You need one solution and a way to make more that keeps the parity right.
4. Find all classes of $x^2 - 3y^2 = 13$ with Theorem 4. There is more than one.
5. Find the fundamental solution of $x^2 - 34y^2 = 1$. If the negative equation had a solution, Theorem 3 would make ε a square.
6. Put $(x_0, y_0) = (1, 0)$ and use $\varepsilon^m = \varepsilon^{m-n}\varepsilon^n$ to run the Euclidean algorithm on the indices. For the second part, find the indices n with $8 \mid y_n$ and with $25 \mid y_n$ separately.

5 Solutions

Problem 1

Find the fundamental solution of $x^2 - 11y^2 = 1$, and the next two solutions in positive integers.

For $y = 1$ and $y = 2$, $11y^2 + 1$ is 12 and 45, neither a square. For $y = 3$ the value is $100 = 10^2$. So the fundamental solution is $(10, 3)$ and $\varepsilon = 10 + 3\sqrt{11}$. Then $\varepsilon^2 = 100 + 60\sqrt{11} + 99 = 199 + 60\sqrt{11}$, and $\varepsilon^3 = (199 + 60\sqrt{11})(10 + 3\sqrt{11}) = 1990 + 1980 + (597 + 600)\sqrt{11} = 3970 + 1197\sqrt{11}$. By Theorem 2 these are the next two solutions, namely $(199, 60)$ and $(3970, 1197)$. As a check, $199^2 - 11 \cdot 60^2 = 39601 - 39600 = 1$. (Note that $11 = 3^2 + 2$, so the formula $(m^2 + 1, m)$ with $m = 3$ also gives $(10, 3)$.)

Answer: $(10, 3)$, then $(199, 60)$ and $(3970, 1197)$

Problem 2

Find the four smallest positive integers n for which $n + 1$ and $5n + 1$ are both perfect squares.

Write $n + 1 = a^2$ and $5n + 1 = b^2$ with a, b positive. Then $b^2 - 5a^2 = -4$, and conversely every positive solution with $a \geq 2$ gives $n = a^2 - 1 \geq 1$ with $5n + 1 = 5a^2 - 4 = b^2$. The fundamental solution of $x^2 - 5y^2 = 1$ is $(9, 4)$, so $\varepsilon = 9 + 4\sqrt{5} \approx 17.94$. By Theorem 4 every positive solution of $b^2 - 5a^2 = -4$ is $v\varepsilon^k$ with $k \geq 0$ and $v = b' + a'\sqrt{5}$ in the window $[2, 2\varepsilon) \approx [2, 35.9)$, where $|b'| \leq 18$ and $|a'| \leq 8$. Checking $a' = 0, \pm 1, \ldots, \pm 8$ for which $5a'^2 - 4$ is a square gives $(b', a') = (\pm 1, \pm 1)$, $(\pm 4, \pm 2)$, $(\pm 11, \pm 5)$, and the ones in the window are $1 + \sqrt{5} \approx 3.24$, $4 + 2\sqrt{5} \approx 8.47$ and $11 + 5\sqrt{5} \approx 22.2$.

So the positive solutions are these three times ε^k, $k \geq 0$. Multiplying by ε gives $29 + 13\sqrt{5}$, $76 + 34\sqrt{5}$ and $199 + 89\sqrt{5}$, and further powers give $a > 89$. So the values of a in increasing order are $1, 2, 5, 13, 34, 89, \ldots$ (every other Fibonacci number). Discarding $a = 1$, which gives $n = 0$, the four smallest values are $n = 2^2 - 1 = 3$, $5^2 - 1 = 24$, $13^2 - 1 = 168$ and $34^2 - 1 = 1155$. For example $5 \cdot 1155 + 1 = 5776 = 76^2$.

Answer: 3, 24, 168, 1155

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Problem 3

Prove that there are infinitely many positive integers n such that n^2+n-1 is five times a perfect square.

We have $n^2+n-1=5m^2$ if and only if $4n^2+4n-4=20m^2$, that is $(2n+1)^2-20m^2=5$. So we only need to find infinitely many solutions of $X^2-20m^2=5$ in positive integers with X odd and $X\geq 3$, since then $n=\frac{X-1}{2}$ is a positive integer with $n^2+n-1=5m^2$.

The pair $(X,m)=(5,1)$ is a solution, and $(9,2)$ solves $x^2-20y^2=1$ because $81-80=1$. Define $(X_0,m_0)=(5,1)$ and $X_{k+1}+m_{k+1}\sqrt{20}=(X_k+m_k\sqrt{20})(9+2\sqrt{20})$, that is

$$X_{k+1}=9X_k+40m_k, \qquad m_{k+1}=2X_k+9m_k.$$

By Brahmagupta's identity each pair satisfies $X_k^2-20m_k^2=5\cdot 1=5$. By induction X_k and m_k are positive and $X_{k+1}>X_k$, so the pairs are distinct. Also $X_{k+1}\equiv 9X_k\equiv X_k \pmod 2$, so every X_k is odd. Hence $n_k=\frac{X_k-1}{2}$ gives infinitely many positive integers with the required property. The first three are $n=2, 42$ and 762. For instance $42^2+42-1=1805=5\cdot 19^2$.

Problem 4

Find all positive integers y less than 1000 for which $3y^2+13$ is a perfect square.

We need the positive solutions of $x^2-3y^2=13$. Here $\varepsilon=2+\sqrt{3}$, and by Theorem 4 every positive solution is $v\varepsilon^k$ with $k\geq 0$ and $v=a+b\sqrt{3}$ a solution with $\sqrt{13}\leq v<(2+\sqrt{3})\sqrt{13}$, roughly $[3.61, 13.46)$, and $|a|<\frac{(3+\sqrt{3})\sqrt{13}}{2}\approx 8.5$, $|b|<4.93$. For $b=0,\pm1,\pm2,\pm3,\pm4$, the number $3b^2+13$ is $13, 16, 25, 40, 61$, so the solutions in range are $(\pm4,\pm1)$ and $(\pm5,\pm2)$. Of these, only $4+\sqrt{3}\approx 5.73$ and $5+2\sqrt{3}\approx 8.46$ lie in the window ($4-\sqrt{3}$ and $5-2\sqrt{3}$ are below 3.61, and the others are negative). So there are two classes.

Multiplying repeatedly by $2+\sqrt{3}$:

$$4+\sqrt{3},\ 11+6\sqrt{3},\ 40+23\sqrt{3},\ 149+86\sqrt{3},\ 556+321\sqrt{3},\ 2075+1198\sqrt{3},$$

$$5+2\sqrt{3},\ 16+9\sqrt{3},\ 59+34\sqrt{3},\ 220+127\sqrt{3},\ 821+474\sqrt{3},\ 3064+1769\sqrt{3}.$$

The y-coordinates increase along each class, so the values below 1000 are $1, 6, 23, 86, 321$ and $2, 9, 34, 127, 474$. In increasing order they are $1, 2, 6, 9, 23, 34, 86, 127, 321, 474$, ten values. For example $3\cdot 474^2+13=674041=821^2$.

Answer: $1, 2, 6, 9, 23, 34, 86, 127, 321, 474$

Problem 5

Prove that the equation $x^2-34y^2=-1$ has no solutions in integers, even though the congruence $x^2\equiv -1 \pmod{34}$ has solutions.

First, $13^2=169=5\cdot 34-1$, so $x^2\equiv -1 \pmod{34}$ is solvable, and no congruence condition of the kind in the lesson rules the equation out.

For $y=1,2,3,4,5$ the number $34y^2+1$ is $35, 137, 307, 545, 851$, none a square (they lie between consecutive squares, as $5^2<35<6^2$, $11^2<137<12^2$, $17^2<307<18^2$, $23^2<545<24^2$, $29^2<851<30^2$), and for $y=6$ the number is $1225=35^2$. So the fundamental solution of $x^2-34y^2=1$ is $(35,6)$ and $\varepsilon=35+6\sqrt{34}$.

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Suppose $x^2 - 34y^2 = -1$ has a solution. Then by Theorem 3 the number η defined there has norm -1, and $\eta^2 = \varepsilon$. Also $\eta > 1$, so $\eta = a + b\sqrt{34}$ with a, b positive integers (shown in the proof of Theorem 3). Comparing coefficients in $(a + b\sqrt{34})^2 = 35 + 6\sqrt{34}$ gives $2ab = 6$ and $a^2 + 34b^2 = 35$. From $ab = 3$ we have $(a, b) = (1, 3)$ or $(3, 1)$, giving $a^2 + 34b^2 = 307$ or 43, not 35. This contradiction shows that there are no solutions.

Problem 6

Let (x_k, y_k), $k = 1, 2, 3, \ldots$, be the solutions of $x^2 - 2y^2 = 1$ in positive integers, in increasing order, so $(x_1, y_1) = (3, 2)$ and $(x_2, y_2) = (17, 12)$. Prove that $\gcd(y_m, y_n) = y_{\gcd(m,n)}$ for all $m, n \geq 1$, and find the smallest n for which 200 divides y_n.

Let $\varepsilon = 3 + 2\sqrt{2}$, the fundamental solution, so $x_k + y_k\sqrt{2} = \varepsilon^k$ for $k \geq 1$ by Theorem 2. Put $(x_0, y_0) = (1, 0)$ so that this also holds for $k = 0$. Since $x_k^2 - 2y_k^2 = 1$, any common divisor of x_k and y_k divides 1, so $\gcd(x_k, y_k) = 1$. For $m \geq n \geq 0$, comparing coefficients of $\sqrt{2}$ in $\varepsilon^m = \varepsilon^{m-n}\varepsilon^n$ gives

$$y_m = x_{m-n}y_n + y_{m-n}x_n.$$

Hence $\gcd(y_m, y_n) = \gcd(y_{m-n}x_n, y_n) = \gcd(y_{m-n}, y_n)$, the last step because $\gcd(x_n, y_n) = 1$. So $\gcd(y_m, y_n)$ is unchanged when the larger index is replaced by the difference of the indices. Running the Euclidean algorithm on (m, n) ends at $(\gcd(m, n), 0)$, so $\gcd(y_m, y_n) = \gcd(y_{\gcd(m,n)}, y_0) = y_{\gcd(m,n)}$, since $y_0 = 0$.

A consequence is that for any positive integer M, the set $S_M = \{n \geq 1 : M \mid y_n\}$ is closed under taking gcds, and the set contains every multiple of each of its elements because $y_r \mid y_{jr}$ (as $\gcd(y_r, y_{jr}) = y_r$). So if S_M is non-empty, the set is the set of multiples of its least element r_M.

The recurrence $y_{k+1} = 2x_k + 3y_k$, $x_{k+1} = 3x_k + 4y_k$ gives $y_1, \ldots, y_4 = 2, 12, 70, 408$. Since $8 \nmid 2, 12, 70$ and $408 = 8 \cdot 51$, we get $r_8 = 4$. Since $5 \nmid 2, 12$ and $5 \mid 70$, $r_5 = 3$, and r_{25} is a multiple of 3 because $S_{25} \subseteq S_5$. Let $z_j = y_{3j}$. As $\varepsilon^3 = 99 + 70\sqrt{2}$ has trace 198 and norm 1, the sequence satisfies $z_{j+1} = 198z_j - z_{j-1}$, with $z_0 = 0$, $z_1 = 70$. Modulo 25, $198 \equiv -2$, and the residues are $z_0, \ldots, z_5 \equiv 0, 20, 10, 10, 20, 0$. So $r_{25} = 15$. Finally $200 \mid y_n$ if and only if $4 \mid n$ and $15 \mid n$, so the smallest such n is 60.

Answer: $n = 60$